



Sentiment & Urgency Detector **AI-Powered Customer Support Ticket Analysis**

- Automated NLP-Driven Ticket Prioritization & Sentiment Classification

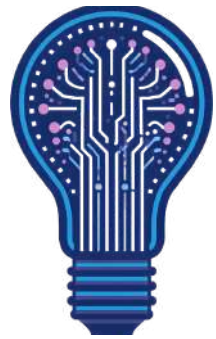
Team Members:

Leela | Mohan | Lalitha | Sunitha



Agenda

Sentiment & Urgency Detector



Introduction



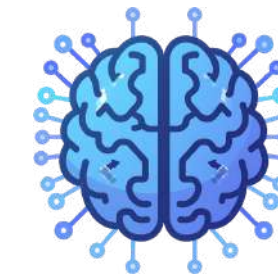
Project Overview



Problem Statement



Existing System



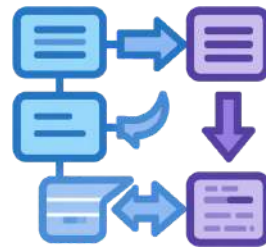
Proposed Solution



Objectives



System
Architecture



Workflow



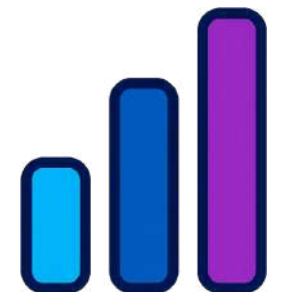
Technologies Used



Key Features



Live Demonstration



Results & Analysis



Introduction



Understanding the need for AI-powered sentiment & urgency detection in modern customer support systems.

In today's fast-paced digital environment, businesses receive thousands of customer support tickets daily. Manually analyzing the tone, sentiment, and urgency of each ticket is time-consuming and error-prone. This project introduces an AI-powered system that automatically detects sentiment (positive, negative, neutral) and urgency levels in support tickets — enabling faster response times, improved customer satisfaction, and smarter resource allocation.





Overview

Explore the end-to-end system design including data ingestion pipelines, AI model layers, backend services, and frontend dashboard components that power the Sentiment & Urgency Detector.



Step-by-step walkthrough of how support tickets are received, preprocessed, analyzed using NLP models, classified by sentiment and urgency, and routed to the appropriate support teams.



Overview of the core technologies powering the solution: Python, TensorFlow, BERT-based NLP models, REST APIs, cloud infrastructure, and interactive analytics dashboards.



The Problem

70% of support tickets are misrouted or delayed due to lack of automated urgency detection.

Customer support teams are overwhelmed by high ticket volumes, relying on manual processes that lead to slow response times, missed urgent issues, and poor customer satisfaction.



Sentiment & Urgency Detector



Delayed Responses

Manual ticket review causes significant delays, leaving urgent customer issues unresolved for hours or even days, impacting satisfaction and retention.



Manual Ticket Sorting

Support agents spend excessive time manually categorizing and prioritizing tickets, reducing efficiency and increasing the risk of human error.



Missed Urgent Issues

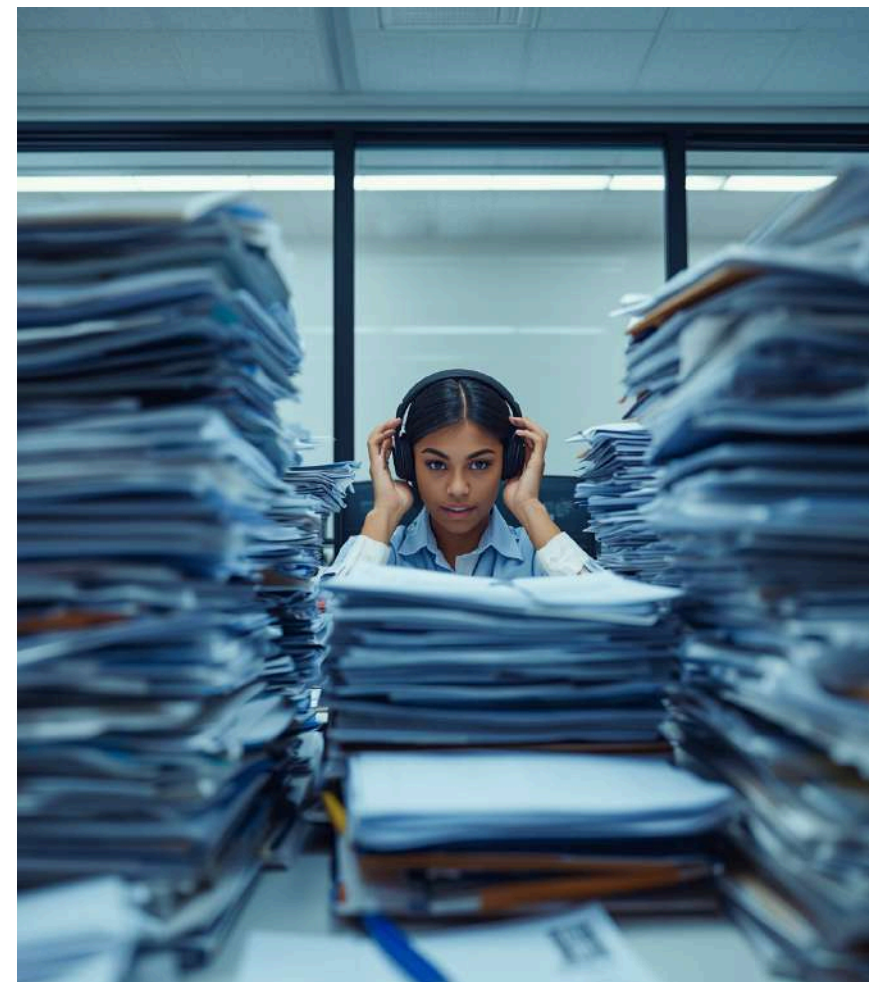
Critical high-priority tickets are frequently buried under low-urgency requests, causing escalations, SLA breaches, and dissatisfied customers.



Existing System

Traditional customer support systems rely on manual ticket triage and classification. Support agents read each ticket individually to assess priority and sentiment, leading to:

- Slow response times due to manual categorization
- Inconsistent urgency assessment across agents
- High volume of tickets causing agent burnout
- No automated sentiment detection or analytics
- Delayed escalation of critical customer issues
- Lack of real-time insights and reporting dashboards





Proposed Solution



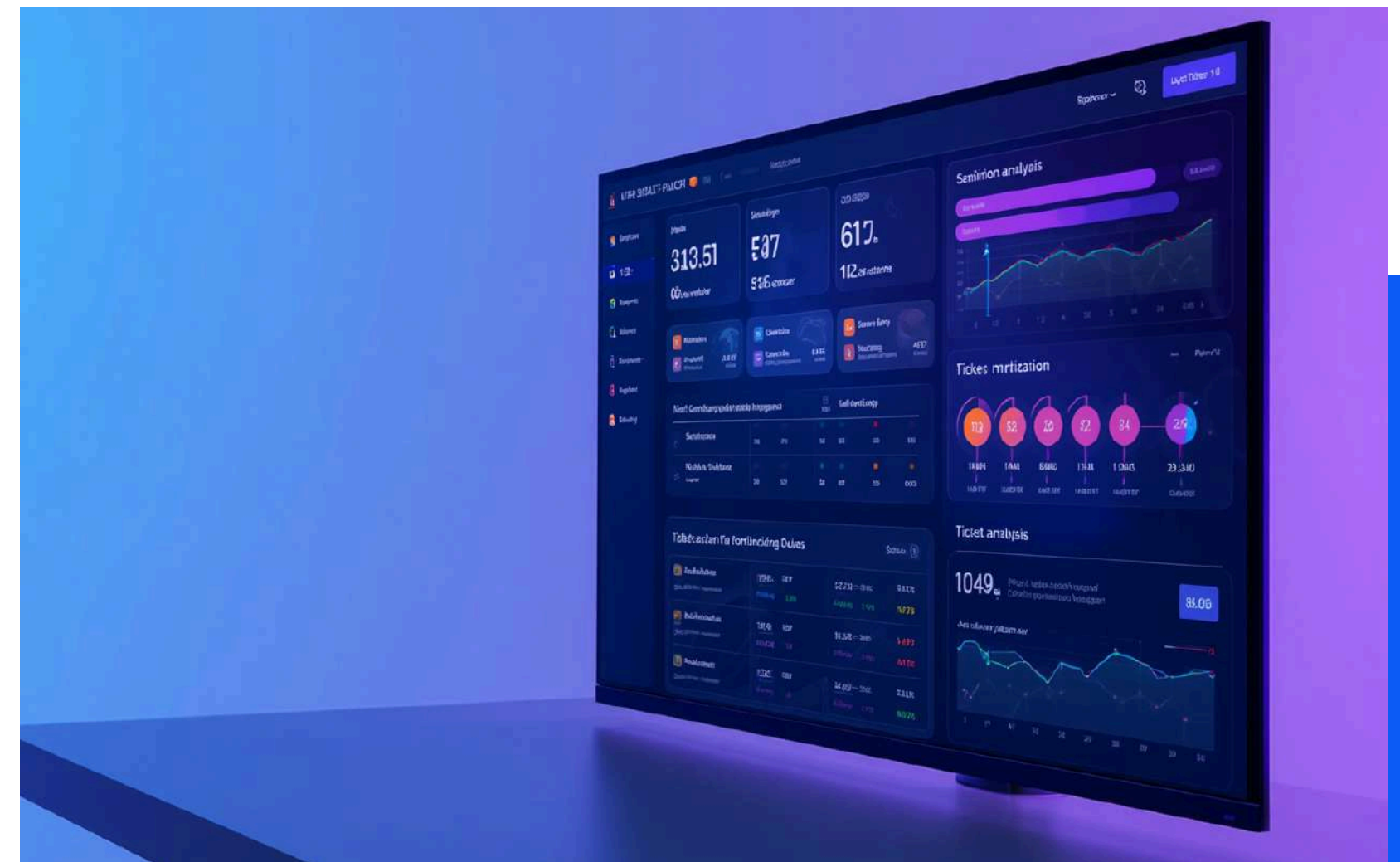
AI-Powered Sentiment Detection

Leverage Natural Language Processing (NLP) and machine learning models to automatically analyze customer support tickets, detecting emotional tone (positive, negative, neutral) and assigning sentiment scores in real time.



Automated Urgency Prioritization

Classify incoming tickets by urgency level (Critical, High, Medium, Low) using AI-driven analysis of keywords, sentiment intensity, and historical patterns — enabling support teams to respond faster and reduce resolution time.

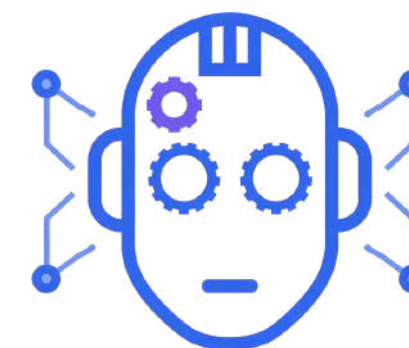




Objectives

- Automate sentiment analysis of support tickets
- Detect urgency levels for faster prioritization
- Improve agent response times & efficiency

Reduce manual effort in ticket classification.
Goal: Enhance customer satisfaction through AI-driven intelligent support workflows.

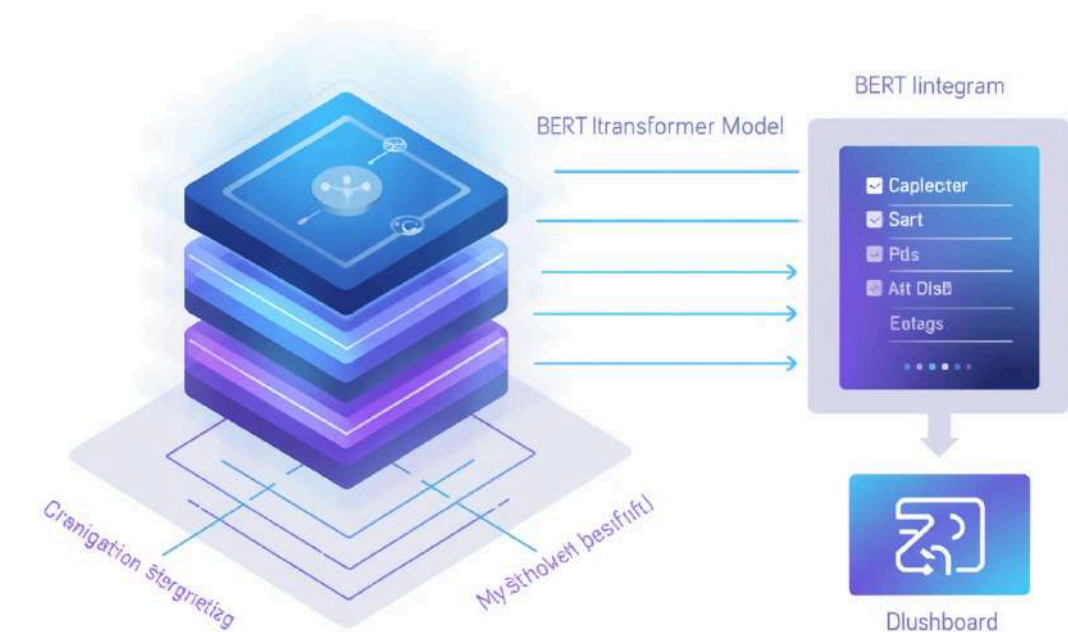
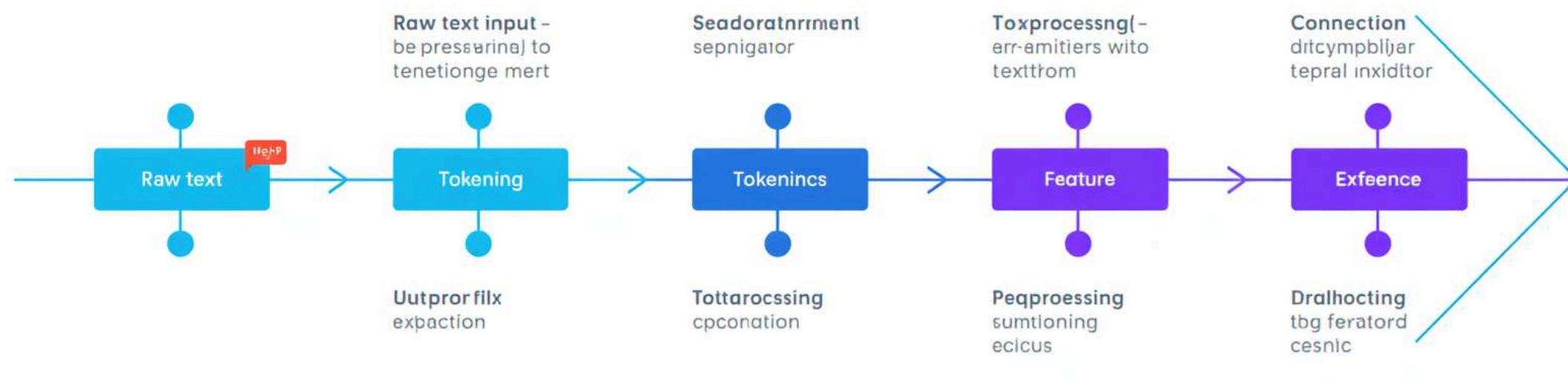




Sentiment & Urgency Detector

System Architecture

The system follows a layered NLP pipeline: raw support tickets are ingested and preprocessed (tokenization, stopwords removal, lemmatization), then passed through ML models (BERT-based transformer + Random Forest) for sentiment classification (Positive/Neutral/Negative) and urgency scoring (Low/Medium/High). Results are stored in a structured database and surfaced via a real-time analytics dashboard, enabling automated ticket routing and priority escalation through REST API integration with CRM platforms.





Workflow

STEP 1: TICKET INGESTION & PREPROCESSING

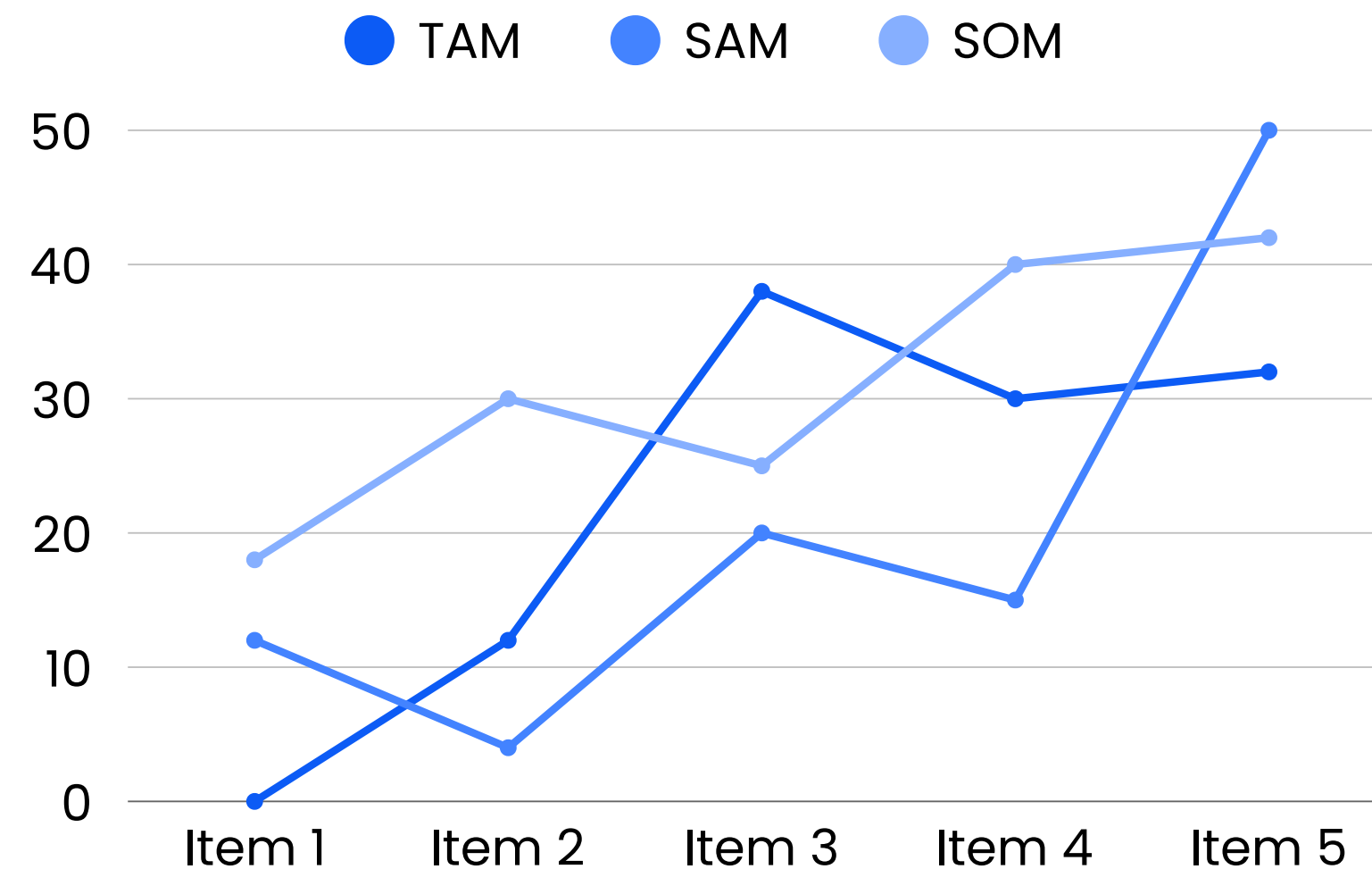
Customer support tickets are collected from multiple channels (email, chat, CRM). Raw text is cleaned, tokenized, and normalized. Duplicate and spam tickets are filtered out before passing to the analysis pipeline.

STEP 2: SENTIMENT ANALYSIS & URGENCY SCORING

NLP models analyze ticket text to detect sentiment (positive, neutral, negative). An urgency scoring engine evaluates keywords, tone, and context to assign a numeric urgency score from 1–10 for each ticket.

STEP 3: PRIORITY ASSIGNMENT & ROUTING

Tickets are automatically classified into priority tiers (Critical, High, Medium, Low) based on combined sentiment and urgency scores. High-priority tickets are instantly routed to the appropriate support team for rapid resolution.





Technologies Used

Built with Python, TensorFlow, and NLP libraries. Cloud services power scalability, while interactive dashboard frameworks deliver real-time insights and analytics.





Key Highlights & Closing



Advantages

Improved response times, automated ticket prioritization, reduced agent workload, higher customer satisfaction scores, and scalable AI-driven support operations.



Limitations & Future Enhancements

Current constraints include multilingual support gaps and domain-specific training needs. Future plans include real-time learning, voice ticket analysis, and CRM integration.



Conclusion

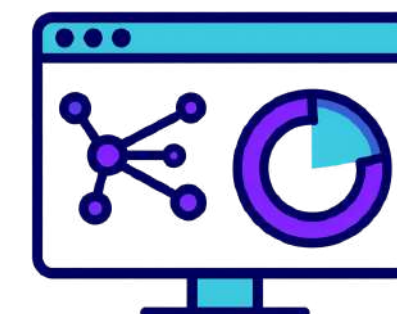
The Sentiment & Urgency Detector delivers intelligent, AI-powered ticket analysis that transforms customer support efficiency through NLP, automation, and actionable insights.



Key Features

- Real-time sentiment & urgency analysis of tickets
- Multi-language support with NLP processing
- Interactive dashboard with live visualizations

Seamless REST API integration with support platforms.
Auto-prioritization, trend analytics & alert notifications.



NLP Support
Multi-Language, weinier supports



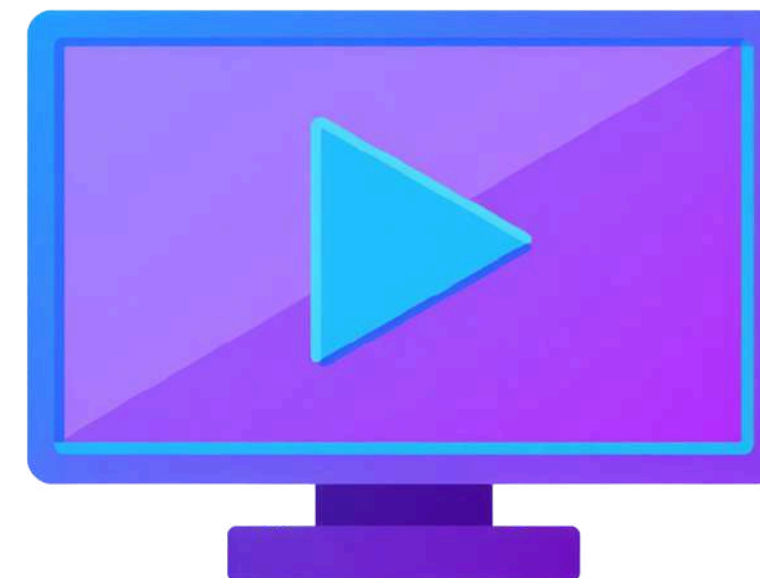
Live Demonstration

input

the product stopped working yesterday and i need immediate assistance

Konathala

A walkthrough of the live system showcasing end-to-end ticket analysis, real-time dashboard updates, sentiment scoring, and urgency flagging using sample customer support data.



Sentiment

Negative

Urgency

High

Priority

Critical Complaint

Sentiment & Urgency Detector

Enter Customer Complaint

I am frustrated and nobody helped me

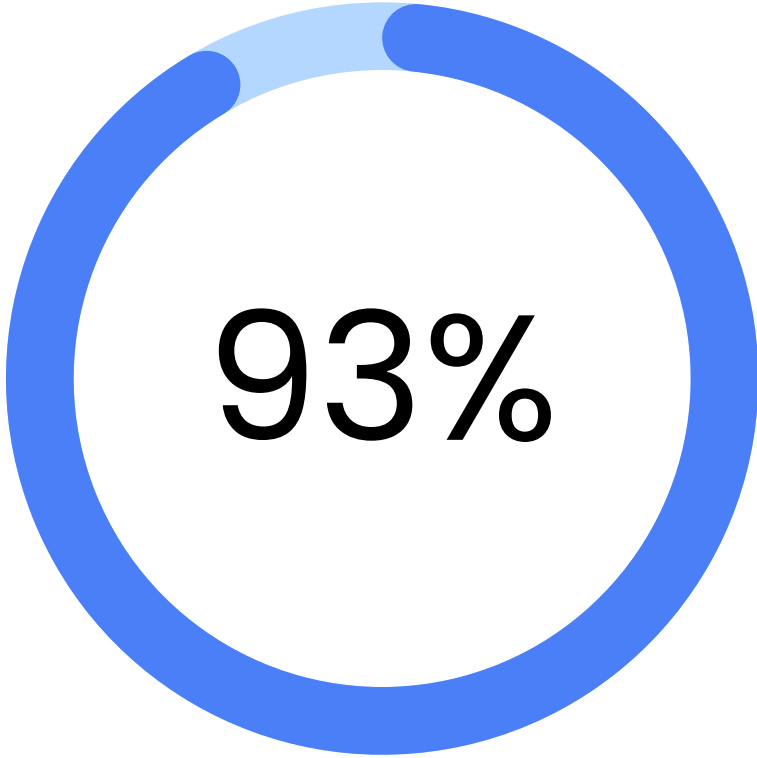
Analyze

Result

Sentiment: Angry / Negative

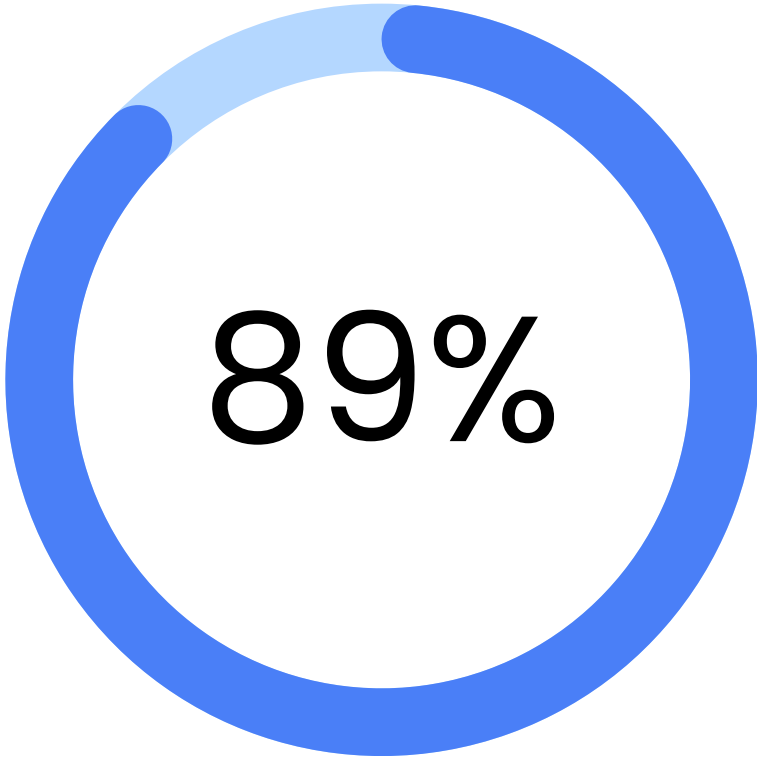
Urgency: High

Results & Analysis



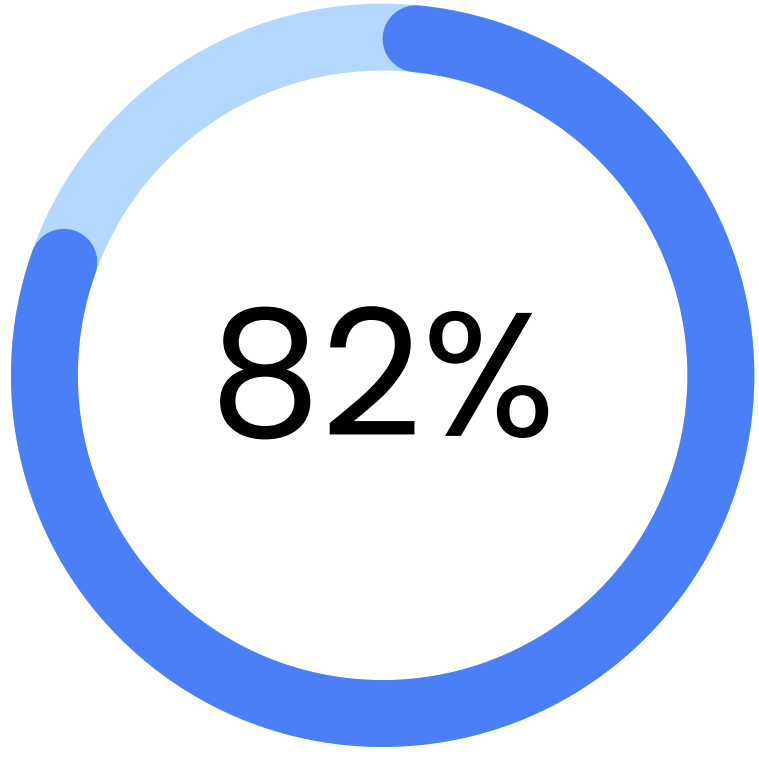
Sentiment Accuracy

Correctly classifies positive, neutral, and negative customer messages.



Urgency Detection

Accurately flags high-priority tickets requiring immediate attention



Hurn Prediction

Identifies at-risk customers before they disengage.

The system was validated on real-world customer support datasets. Sample outputs demonstrate accurate classification across sentiment categories, urgency levels, and churn-risk indicators — with the recommendation engine providing actionable next steps for each ticket.

Future Enhancements



Discord Alert Example

text



High-Risk Customer Ticket

Risk Level: HIGH

Sentiment Score: 0.94

Urgency Score: 0.91

Churn Risk Score: 0.88

Reasons:

- Strong negative language
- Immediate action requested
 - Churn intent detected

Ticket:

"Your service has been down for two days..."

- Multi-language support
- Voice-based ticket analysis
- Advanced deep learning models
- CRM and helpdesk integration
- Mobile application



Thank You!

